Vishal Raj P R

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About Me

AI/ML Engineer and Python Developer with experience building intelligent, real-time systems using machine learning, voice AI, computer vision, and full-stack web. Skilled in LLM orchestration, RAG pipelines, and multi-agent AI architectures. Adept at shipping deployable solutions end to end and integrating hardware with software to solve real-world problems.

Education

Hindustan Institute of Technology and Science — 2023 – 2027
Bachelor of Technology in Computer Science — *CGPA: 8.01*

St. John's Public School (CBSE) — 2022 – 2023
Senior Secondary — *10th: 67.5%* *12th: 74.6%*

Technical Skills

Languages: Python, SQL, JavaScript, TypeScript
AI & ML: Machine Learning, SHAP, PyTorch, NLP, Speech Recognition (Whisper, Silero VAD), TTS
LLM & Agentic AI: LangChain, Multi-Agent Systems, RAG, Vector Databases (Pinecone), Prompt Engineering
Frameworks & Tools: Streamlit, Tkinter, Arduino (IoT), Git, GitHub

Work Experience

AI Developer Intern — Jun 2026 – Oct 2026
Lexbolt — *Remote, London, UK*

- Prepared, annotated, and validated datasets and fine-tuned AI/ML models for Lexbolt's Regulations Intelligence Module to improve regulatory information-extraction accuracy.
- Built data pipelines to ingest and process regulatory documents; tested model outputs before production integration, using Git for version-controlled model iteration.

Projects

HITS CareerHub – Placement Platform (Web + Android) | *Node.js, PostgreSQL, React Native, Gemini* | GitHub

- Built and deployed a full-stack placement platform (about 200 REST endpoints, 51-table schema, Google SSO, four-tier role-based access) replacing spreadsheet-based recruitment, with eligibility rules, drive pipelines, proctored online tests, QR attendance, and a React Native Android app.
- Added a Gemini function-calling placement assistant grounded in real data lookups, and fixed every high-severity finding from a security audit (XSS, open redirects, session revocation, CSV injection).

JARVIS – Real-Time Voice Assistant for Calls | *Python, FastAPI, WebSockets, Whisper, PostgreSQL* | GitHub

- Built and deployed a voice assistant (Whisper STT, Silero VAD, LLM tool calling, Edge-TTS) that joins live calls, handles Tamil-English code-switched speech, and ships as a PWA and Android app; designed a three-gate privacy model as the basis of a research paper.
- Fixed a production outage caused by a polling loop exhausting the database quota, then migrated Postgres providers.

Parkinson's Disease Detection | *Python, Gradient Boosting, SHAP, Scikit-learn* | GitHub

- Developed a Gradient Boosting classifier on voice biomarker data achieving **84.23% accuracy**; applied **SHAP** to quantify each vocal attribute's contribution. Won **Best Paper Award** at the International Conference on Quantum Horizons 2025.

AI Voice Assistant for the Blind | *Python, Computer Vision, OCR, TTS* | GitHub

- Built a real-time assistive system using computer vision and OCR to convert live camera feeds and on-screen text into speech via TTS, with voice commands and an accessible GUI.

Antitheft Footstep Detection System | *Arduino, GSM, Embedded Systems, IoT* | GitHub

- Built an Arduino and analog pressure sensor system that detects unauthorized footsteps, sounds a buzzer, and sends an SMS alert via a SIM800L GSM module, with a configurable threshold and 8-second cooldown to prevent false positives.

Achievements & Certifications

- **Best Paper Award** – International Conference on Quantum Horizons 2025 (HITS & KCG College of Technology, Chennai, 18–19 Dec 2025), for *Intelligent Speech Signal Analysis for Early Detection of Parkinson's.*
- **The AI Engineer Course 2026: AI Engineer Bootcamp** – Udemy (365 Careers), Jun 2026 (29.5 hrs). Certificate
- **Machine Learning** – Internshala: supervised learning, model evaluation, and hands-on data applications. Certificate
- **Python & SQL** – Greens Technology: practical training in Python and SQL with real-world database projects.